MATHEMATICAL RESEARCH · AMSTERDAM · 2026

# Linear Diophantine Representation Systems

## $N = 19A + 9B$

*$N = pA + qB$ with $p \equiv 1 \pmod q$*

*A Framework of Modular Structure — Made Visible Through Representations*

Digital Root · Representations · Complete Family · Perfect Numbers
Fibonacci · Symmetry Points · Additivity Theory · Matryoshka Theory


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> *"The focus lies not on the numbers themselves, but on their underlying structure. A number is not a magical or golden object — it is a reference point within a perfectly calibrated system."*

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ABSTRACT

This document develops a complete theory of linear Diophantine systems $N = pA + qB$, where $p$ and $q$ are coprime positive integers satisfying $p \equiv 1 \pmod q$. The central result is that the minimal coefficient $A_0$ always equals the generalised digital root $\mathrm{dr}_q(N) = 1 + ((N-1) \bmod q)$, provided $N \geq pq$.

A closed formula for the number of representations $R(p,q;N)$ is derived, the threshold values and family boundary values are characterised, and it is proved that the step constant equals $pq$.

Three structural theorems form the core of the theory:

• **Main Theorem** — $A_0 = \mathrm{dr}(N)$ for every $N \geq pq$

• **Additivity Theory** — If $a + b = c$, representations sum simultaneously on three levels

• **Matryoshka Theory** — Every value in the reduction chain $N \to S(N) \to \ldots \to \mathrm{dr}(N)$ appears as an $A$-coordinate in the representations of $N$

## Contents

# §0 Guide — Notation and Worked Examples

This chapter explains every letter and every concept with concrete examples. No prior knowledge is required.

## The basic equation

$$\boxed{N = 19 \times A + 9 \times B}$$

Given a number $N$, we seek all ways to write $N$ as a multiple of 19 plus a multiple of 9, with $A, B \geq 0$.

## Meaning of each letter (Example: N = 958)

| LETTER | NAME | MEANING | EXAMPLE (N=958) |
|---|---|---|---|
| N | The number | The number to write as 19A+9B | 958 |
| A | First coordinate | How many times 19 is used | 4 (19×4=76) |
| B | Second coordinate | How many times 9 is used | 98 (9×98=882) |
| R(N) | Representations | Number of ways to write N | R(958)=6 |
| dr(N) | Digital root | Sum digits until one digit: 9+5+8=22, 2+2=4 | dr(958)=4 |
| S(N) | Digit sum | Sum of all digits of N (once) | S(958)=22 |
| $A_0$ | Smallest A | Always equals dr(N) | $A_0$=4=dr(958) |

## All representations of N = 958

| # | A | B | 19×A | 9×B | SUM | REMARK |
|---|---|---|---|---|---|---|
| 1 | 4 | 98 | 76 | 882 | 958 | |
| 2 | 13 | 79 | 247 | 711 | 958 | |
| 3 | 22 | 60 | 418 | 540 | 958 | |
| 4 | 31 | 41 | 589 | 369 | 958 | 19×31=589 |
| **5 ★** | **40** | **22** | **760** | **198** | **958** | **SYMMETRY POINT: dr(A)=dr(B)=dr(N)=4** |
| 6 | 49 | 3 | 931 | 27 | 958 | |

## Step and boundaries

| CONCEPT | FORMULA | MEANING | EXAMPLE |
|---|---|---|---|
| Step size | `+9 in A, -19 in B` | Each successive representation | `(4,98)`→`(13,79)`→… |
| Step constant | `19×9 = 171` | Distance between families | Every 171 numbers R rises by 1 |
| Frobenius bound | `(19-1)(9-1) = 144` | Every N ≥ 144 has ≥1 representation | `N=144 has R=1` |
| Boundary G(k) | `171(k-1)+144` | Every N ≥ G(k) has ≥k representations | `G(10)=1683` |

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# §1 Introduction and Historical Context

The equation $N = 19A + 9B$ looks simple, but contains an unexpected richness of structures. The choice of (19, 9) is not arbitrary:

- $19 + 9 = 28$ (the second perfect number)
- $\varphi(19) = 18 = 2{\cdot}9$
- $19 \equiv 1 \pmod 9$

These three properties create a deep connection between digital roots, representation counts, and perfect numbers. The primary goal is to generalise the observed patterns to arbitrary pairs $(p, q)$ and to prove that (19, 9) is unique in its combination of five strong properties.

## The three fundamental properties of (19, 9)

| PROPERTY | FORMULA | CONSEQUENCE IN THE SYSTEM |
|---|---|---|
| Congruence | `19 ≡ 1 (mod 9)` | dr(N)=dr(A) for every representation of N |
| Step vector | `A→A+9, B→B-19` | Every successive representation shifts exactly this way |
| Step constant | `19×9 = 171` | Fixed distance between all threshold values |
| Frobenius threshold | `(19-1)(9-1) = 144` | Every N ≥ 144 has at least one representation |
| Euler key | `φ(19) = 18 = 2×9` | Coupling of the two bases |

## Comparison with the extended Euclidean algorithm

For any pair $(p, q)$ with $\gcd(p, q) = 1$, the classical extended Euclidean algorithm finds one particular solution $(x_0, y_0)$ to $px + qy = N$ and then derives all others through a parameter $t$. The 19-9 system bypasses this multi-step process entirely. Because $p \equiv 1 \pmod{q}$, the minimal $A$-coordinate is simply $A_0 = N \bmod q$, obtainable in a single operation.

| STEP | EXTENDED EUCLID | 19-9 SYSTEM (DIRECT) |
|---|---|---|
| 1 | Compute gcd(p,q) via repeated division | $A_0$ = N mod 9 — one modulo operation |
| 2 | Back-substitution for one solution $(x_0,y_0)$ | Step vector: $A_0$, $A_0+9$, $A_0+18$, ... |
| 3 | Determine parameter t so that x, y ≥ 0 | R(N) = formula — done after step 2 |
| 4 | All solutions via $x=x_0+qt$, $y=y_0-pt$ | — already complete after step 2 |

Example N = 958: Euclid gives the initial solution (958, −1916) — both values negative, requiring further adjustment. Direct: $A_0 = 958 \bmod 9 = 4$, all 6 representations via (+9, −19).

For N greater than roughly 200, the direct formula $A_0$ = N mod 9 becomes considerably more efficient than the extended Euclidean algorithm. For N in the millions or beyond, R(N) ~ N/171 representations are all generated immediately from the step vector, without any iterative computation.

## §2 Foundations and Notation

We consider pairs $(p, q)$ of positive integers with $\gcd(p, q) = 1$, $p \equiv 1 \pmod{q}$, $p > q \geq 1$. For each $N$, $R(p, q; N)$ denotes the number of pairs $(A, B) \in \mathbb{Z}^2_{\geq 0}$ with $pA + qB = N$.

DEFINITION 2.1 — GENERALISED DIGITAL ROOT

$\mathrm{dr}_q(N) = 1 + ((N - 1) \bmod q) \in \{1,\ldots,q\}$.

For $q = 9$ this is the ordinary decimal digital root. □

THEOREM 2.2 — MAIN THEOREM: $A_0$ = DR(N)

For $N \geq pq$, the smallest $A$ for which $N = pA + qB$ has a solution with $B \geq 0$ is equal to $\mathrm{dr}_q(N)$. All solutions are: $A = A_0 + qk$, $B = (N - pA_0)/q - pk$ with $k = 0, 1, \ldots, k_{max}$.

PROOF

From $N = pA + qB$ modulo $q$: $N \equiv pA \equiv A \pmod{q}$ because $p \equiv 1 \pmod{q}$. The smallest non-negative $A$ with that congruence is $A_0 = N \bmod q$. Since $N \geq pq$, we have $B \geq 0$. The step $A \to A+q$, $B \to B-p$ yields all representations. □

**Boundary case N < pq:** If $N \equiv 0 \pmod{q}$, then $A = 0$ is also a valid solution, since $0 \cdot p + (N/q) \cdot q = N$. In this case the representation $(0, N/q)$ must be explicitly included.

## §3 Closed Formula for R(p, q; N)

THEOREM 3.1 — REPRESENTATION FORMULA

For $N \geq pq$ and $r = \mathrm{dr}_q(N)$:

$$\boxed{R(p, q; N) = \lfloor (\lfloor N/p \rfloor - r) / q \rfloor + 1}$$

***Corollary 3.2:*** *$R(p, q; N) \sim N/(pq)$ for large $N$.*

## Verification table

| N | CHAIN | ⌊N/19⌋ | N MOD 9 | COMPUTATION | R(N) |
|---|---|---|---|---|---|
| **366** | **366→15→6** | **19** | **6** | **⌊(19-6)/9⌋+1** | **2** |
| **592** | **592→16→7** | **31** | **7** | **⌊(31-7)/9⌋+1** | **3** |
| **589** | **589→22→4** | **31** | **4** | **⌊(31-4)/9⌋+1** | **4** |
| 369 | 369→18→9 | 19 | 0 | ⌊(19-0)/9⌋+1 | 3 |
| **958** | **958→22→4** | **50** | **4** | **⌊(50-4)/9⌋+1** | **6** |
| 2334 | 2334→12→3 | 122 | 3 | ⌊(122-3)/9⌋+1 | 14 |
| 6666 | 6666→24→6 | 350 | 6 | ⌊(350-6)/9⌋+1 | 39 |

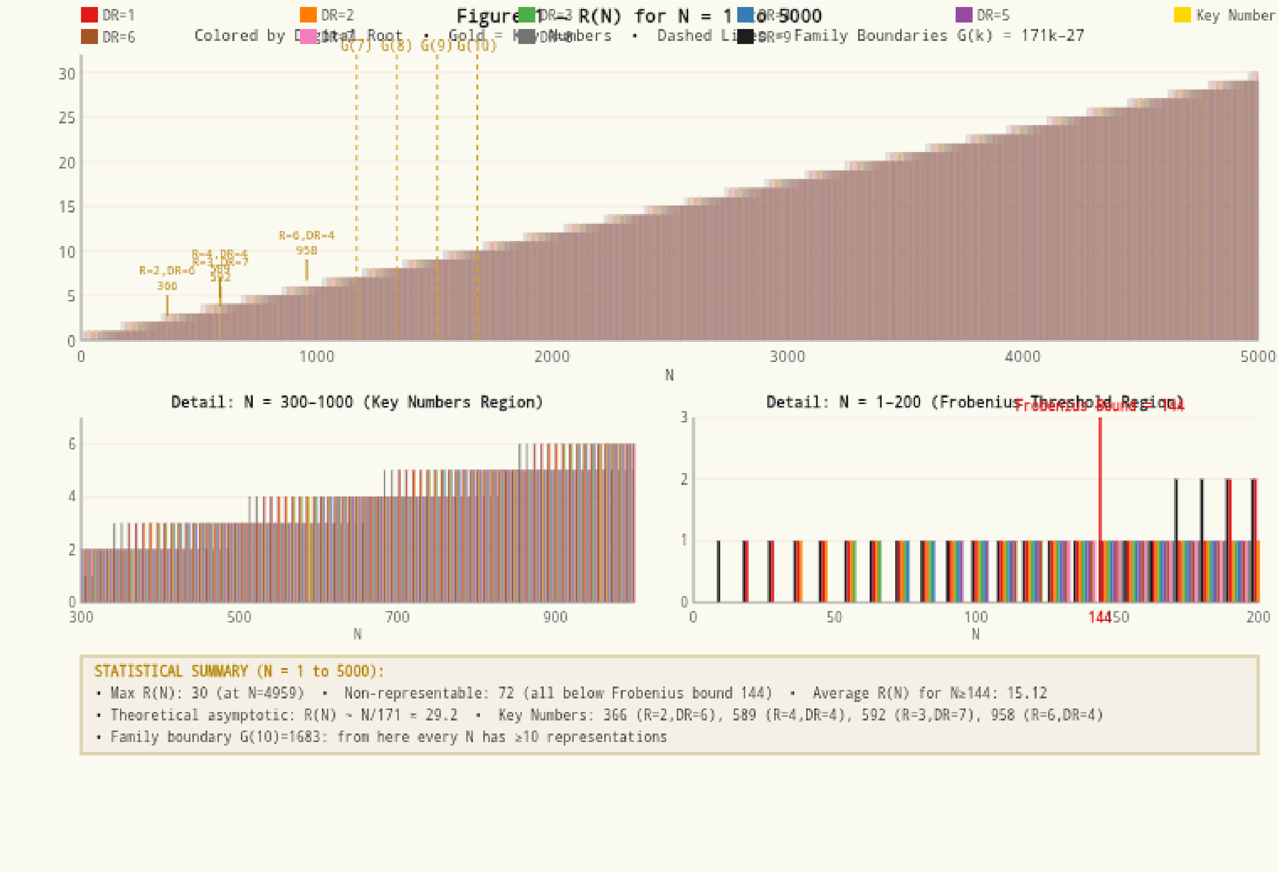


*Figure 1 — R(N) for N=1–5000, coloured by digital root. Key numbers 366, 589, 592, 958 highlighted in gold. Detail views: N=300–1000 (key numbers region) and N=1–200 (Frobenius threshold region). Statistical summary included.*

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## §4 Threshold Values and Family Boundary Values

For a fixed digital root $r$, the $t$-th threshold value is $T(t, r) = 19r + 171(t-1)$. Each time the minimum number of representations rises by 1, the boundary shifts by exactly $171 = 19\times 9$.

$$\boxed{G(k) = 171(k-1) + 144 = 171k - 27}$$

### Table 1: First N per digital root where R(N) = t

| T | DR 1 | DR 2 | DR 3 | DR 4 | DR 5 | DR 6 | DR 7 | DR 8 | DR 9 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 19 | 38 | 57 | 76 | 95 | 114 | 133 | 152 | 9 |
| 2 | 190 | 209 | 228 | 247 | 266 | 285 | 304 | 323 | 171 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 3 | 361 | 380 | 399 | 418 | 437 | 456 | 475 | 494 | 342 |
| 4 | 532 | 551 | 570 | **589** | 608 | 627 | 646 | 665 | 513 |
| 5 | 703 | 722 | 741 | 760 | 779 | 798 | 817 | 836 | 684 |
| 6 | 874 | 893 | 912 | 931 | 950 | 969 | 988 | 1007 | 855 |
| 7 | 1045 | 1064 | 1083 | 1102 | 1121 | 1140 | 1159 | 1178 | 1026 |
| 8 | 1216 | 1235 | 1254 | 1273 | 1292 | 1311 | 1330 | 1349 | 1197 |
| 9 | 1387 | 1406 | 1425 | 1444 | 1463 | 1482 | 1501 | 1520 | 1368 |
| 10 | 1558 | 1577 | 1596 | 1615 | 1634 | 1653 | 1672 | — | 1539 |

*Formula: T(t,r) = 19r + 171(t−1). Rows differ by 171; columns differ by 19. 589 (t=4, DR4) highlighted in gold.*

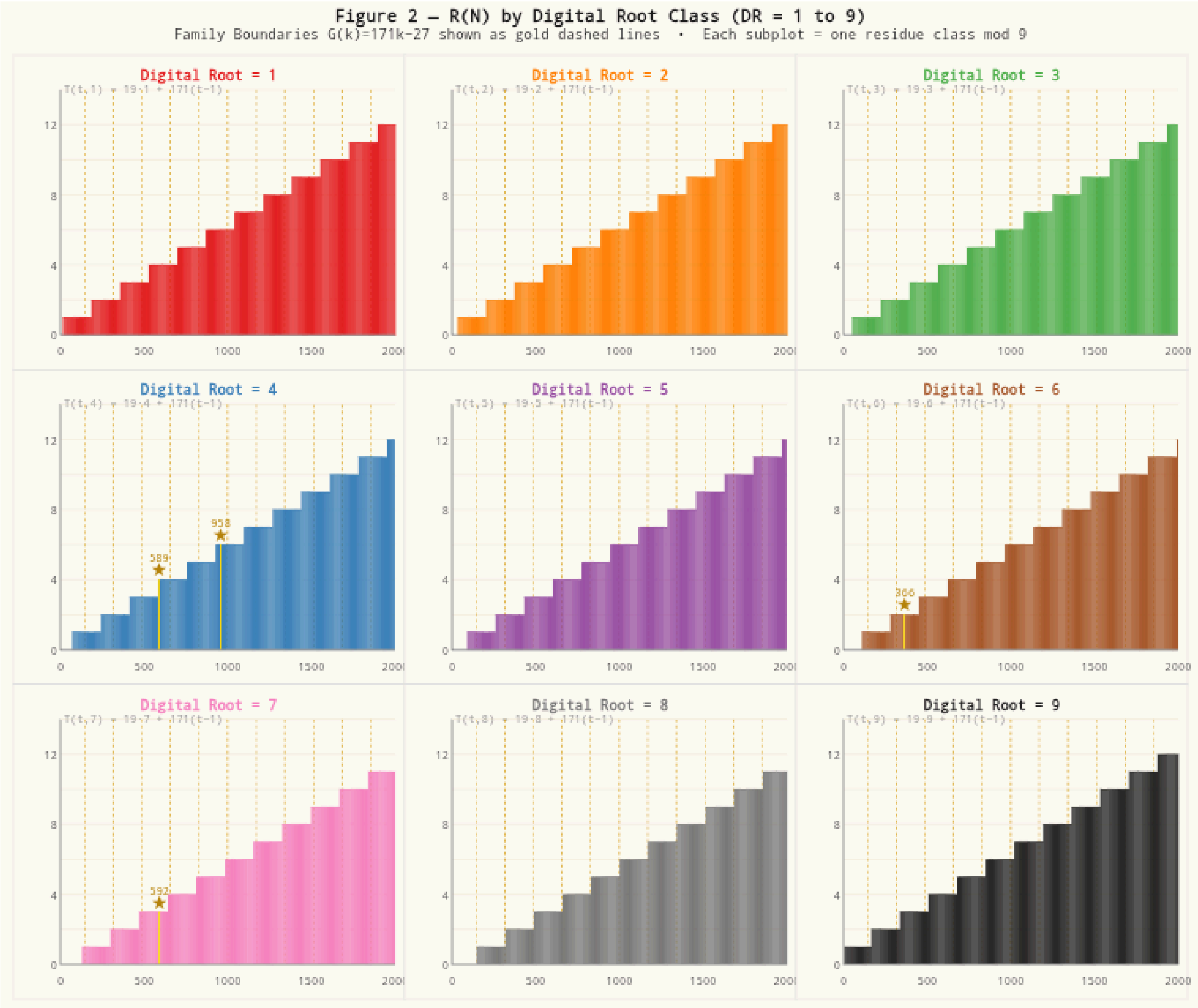


*Figure 2 — R(N) by digital root class (DR=1 to 9), N=1–2000. Each subplot shows one residue class mod 9. Family boundaries G(k) = 171k−27 shown as gold dashed lines. Key numbers marked with gold stars ★.*

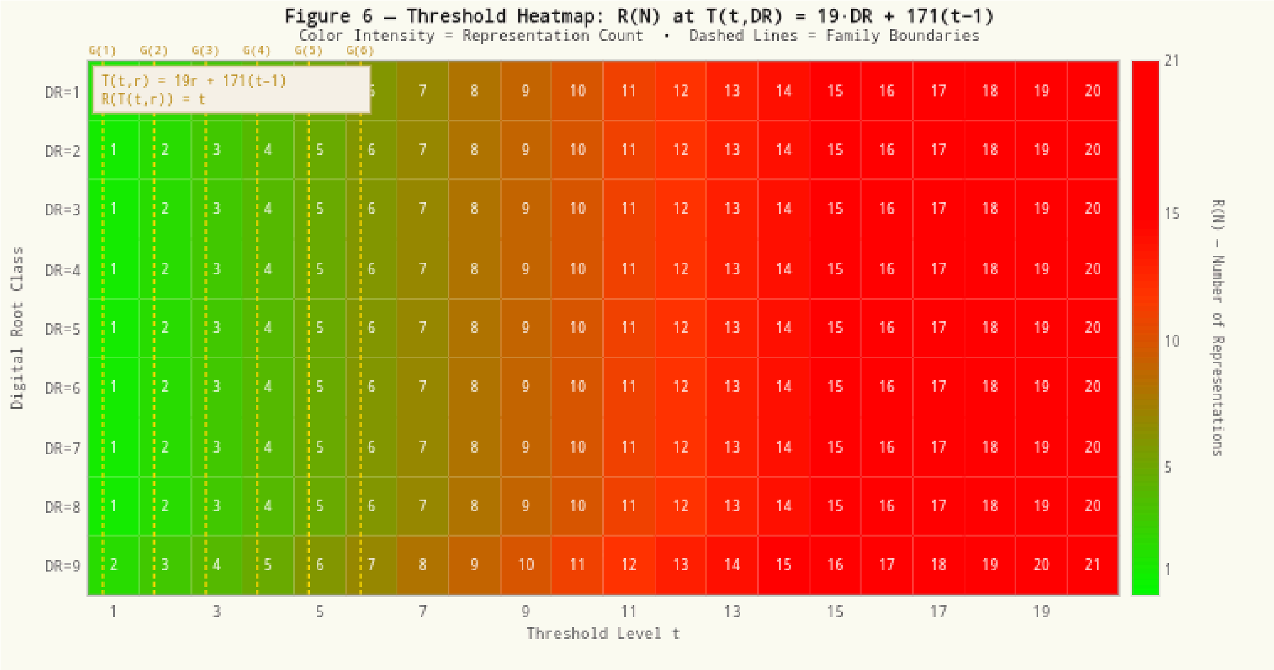


*Figure 6 — Threshold heatmap: R(N) at T(t, DR) = 19·DR + 171(t−1). Colour intensity = representation count. Dashed lines = family boundaries G(k).*

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# §5 Symmetry Points

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**DEFINITION 5.1 — SYMMETRY POINT**

A representation $(A, B)$ of $N$ is a **symmetry point** if $\mathrm{dr}(A) = \mathrm{dr}(B) = \mathrm{dr}(N)$.

## Example: N = 958 (dr = 4)

| # | A | B | 19A | 9B | DR(A) | DR(B) |
|---|---|---|---|---|---|---|
| 1 | 4 | 98 | 76 | 882 | 4 | 8 |
| 2 | 13 | 79 | 247 | 711 | 4 | 7 |
| 3 | 22 | 60 | 418 | 540 | 4 | 6 |
| 4 | 31 | 41 | 589 | 369 | 4 | 5 |
| **5 ★** | **40** | **22** | **760** | **198** | **4** | **4** |

| | | | | | | |
|---|---|---|---|---|---|---|
| 6 | 49 | 3 | 931 | 27 | 4 | 3 |

Further examples:

- N = 1680 (dr = 6): symmetry point (60, 60) where A = B
- N = 6666 (dr = 6): symmetry point (24, 690) — dr(24)=6, dr(690)=dr(15)=6. Verification: 24×19+690×9=456+6210=6666 ✓

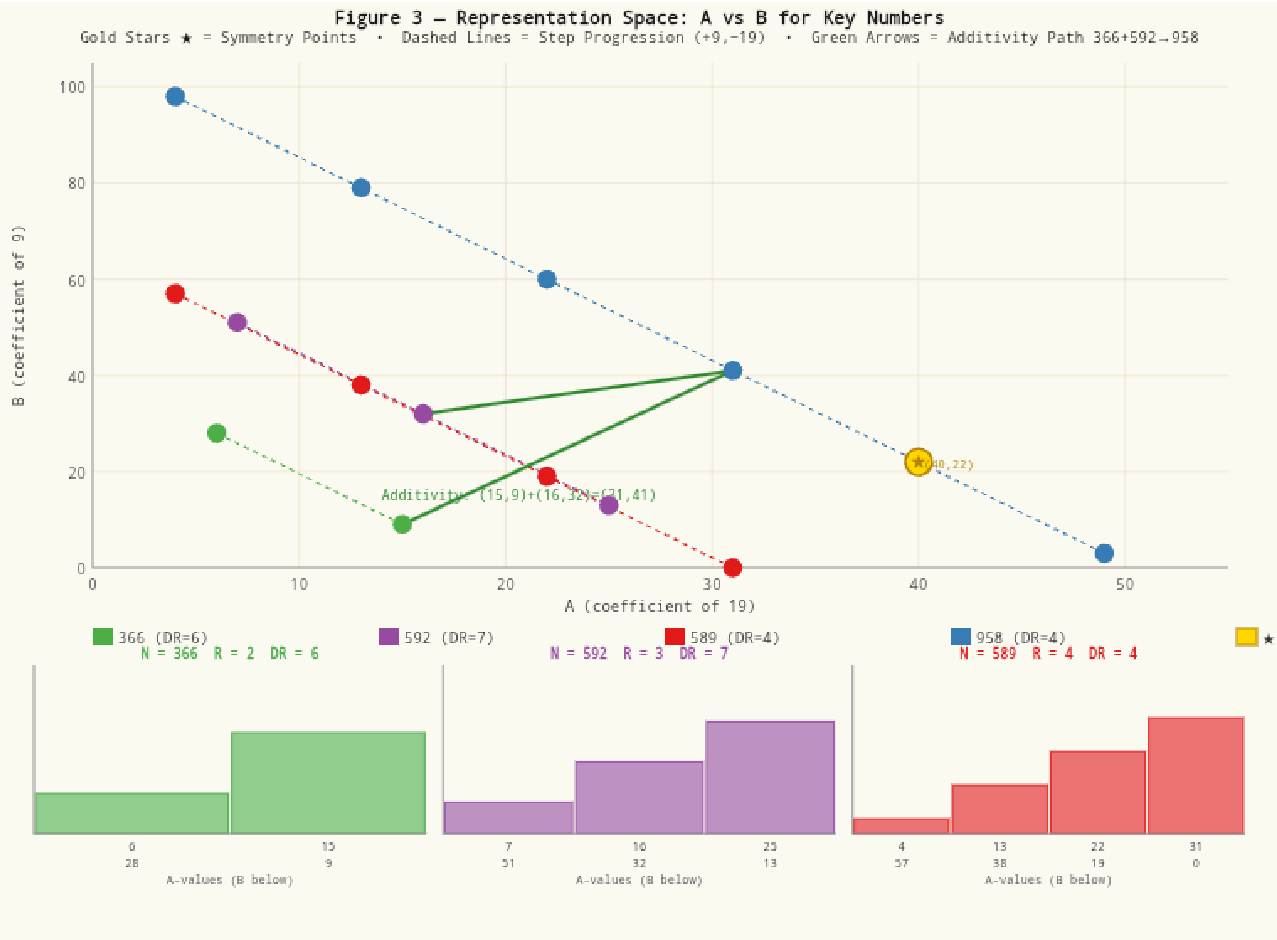


*Figure 3 — Representation space: A vs B for key numbers 366, 592, 589, 958. Gold stars ★ = symmetry points. Dashed lines = step progression (+9, −19). Green arrows = additivity path 366+592=958. Individual representation counts shown below.*

# §6 Reduction Chains — The Central Pattern

The reduction chain of $N$ yields two mandatory $A$-coordinates: the intermediate stop $S(N)$ (first digit sum) and the digital root dr($N$). This holds for every number, without exception.

The reason: the $A$-sequence is $A_0$, $A_0$+9, $A_0$+18, … All elements share the same residue mod 9 as $N$. The intermediate stop $S(N)$ has that same residue, and therefore always falls in this sequence.

| N | CHAIN | REPRESENTATION VIA S(N) | REPRESENTATION VIA DR(N) |
|---|---|---|---|
| **366** | **366→15→6** | **A=15: 15×19+9×9=366 ✓** | **A=6: 6×19+28×9=366 ✓** |
| **592** | **592→16→7** | **A=16: 16×19+32×9=592 ✓** | **A=7: 7×19+51×9=592 ✓** |
| **958** | **958→22→4** | **A=22: 22×19+60×9=958 ✓** | **A=4: 4×19+98×9=958 ✓** |
| 2334 | 2334→12→3 | A=12: 12×19+234×9=2334 ✓ | A=3: 3×19+253×9=2334 ✓ |

**Observation:** The sum 366 + 592 = 958 carries through: intermediate stops 15 + 16 = 31 is an A-coordinate of 958 (31×19+41×9=958). Digital roots 6 + 7 = 13, and dr(13) = 4 = dr(958).

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## §7 Additivity Theory

THEOREM 7.1 — ADDITIVITY

Let $a + b = c$. If $(A_1, B_1)$ is a representation of $a$ and $(A_2, B_2)$ is a representation of $b$, then $(A_1+A_2, B_1+B_2)$ is a valid representation of $c$.

PROOF

$p(A_1+A_2) + q(B_1+B_2) = pA_1+qB_1+pA_2+qB_2 = a+b = c$. □

This holds simultaneously on three levels:

| LEVEL | STATEMENT |
|---|---|
| 1 — Numbers | $a + b = c$ |
| 2 — Coordinates | $(A_1,B_1) + (A_2,B_2)$ is a representation of $c$ |

| | |
|---|---|
| 3 — Digital roots | $dr(a) + dr(b) \equiv dr(c) \pmod 9$ |

## Illustration 1 — 366 + 592 = 958 (Structure Triangle)

| LEVEL | 366 | + 592 | = 958 |
|---|---|---|---|
| Number | 366 | +592 | =958 |
| Min. rep. | (6, 28) | +(7, 51) | =(13, 79) ✓ |
| Alt. rep. | (15, 9) | +(16, 32) | =(31, 41) ✓ |
| Digital root | 6 | +7=13→dr=4 | =dr(958) ✓ |

366 is the only number whose minimal pair (6, 28) consists of the first two perfect numbers.

## Illustration 2 — 589 + 369 = 958 (Anagram Triangle)

589 is an anagram of 958 — the same digits 5, 8, 9 in a different order. Because the digit sum is invariant under permutation: $S(589) = S(958) = 22$ and $dr(589) = dr(958) = 4$. The key lies in the factorisations: $589 = 19{\times}31$ and $369 = 9{\times}41$.

$$589 + 369 = 19{\times}31 + 9{\times}41 = 958$$

### *The complete 4 × 3 = 12 mapping*

Because 589 + 369 = 958, every pair of representations — one from 589, one from 369 — yields via linearity a valid representation of 958. Two of the 12 sums reach the symmetry point (40, 22):

| FROM 589 | FROM 369 | = REPRESENTATION OF 958 |
|---|---|---|
| (4, 57) | (0, 41) | (4, 98) |
| (4, 57) | (9, 22) | (13, 79) |
| (4, 57) | (18, 3) | (22, 60) |
| (13, 38) | (0, 41) | (13, 79) |
| (13, 38) | (9, 22) | (22, 60) |
| (13, 38) | (18, 3) | (31, 41) |

| | | |
|---|---|---|
| (22, 19) | (0, 41) | (22, 60) |
| (22, 19) | (9, 22) | (31, 41) |
| **(22, 19)** | **(18, 3)** | **(40, 22) ★ PATH 1** |
| (31, 0) | (0, 41) | (31, 41) |
| **(31, 0)** | **(9, 22)** | **(40, 22) ★ PATH 2** |
| (31, 0) | (18, 3) | (49, 3) |

## Illustration 3 — 592 + 366 = 958 (Intermediate Stops)

| LEVEL | 592 | + 366 | = 958 |
|---|---|---|---|
| Number | 592 | +366 | =958 |
| Int. stops | 16 | +15=31 | dr(31)=4=dr(958) ✓ |
| Digital roots | 7 | +6=13 | dr(13)=4=dr(958) ✓ |
| Rep. | (16, 32) | +(15, 9) | =(31, 41) ✓ |

## General linearity pattern

The additivity of representations is not a property of these specific numbers. It holds for every triple $a + b = c$, regardless of the values involved.

| $N_1$ | $N_2$ | $N_3$ | $S(N_1)+S(N_2)$ | A-COORD OF $N_3$? | $DR(N_3)$ |
|---|---|---|---|---|---|
| 1000 | 2000 | 3000 | 1+2=3 | ✓ | 3 |
| 333 | 666 | 999 | 9+18=27 | ✓ | 9 |
| 100 | 200 | 300 | 1+2=3 | ✓ | 3 |
| 1234 | 2345 | 3579 | 10+14=24 | ✓ | 6 |

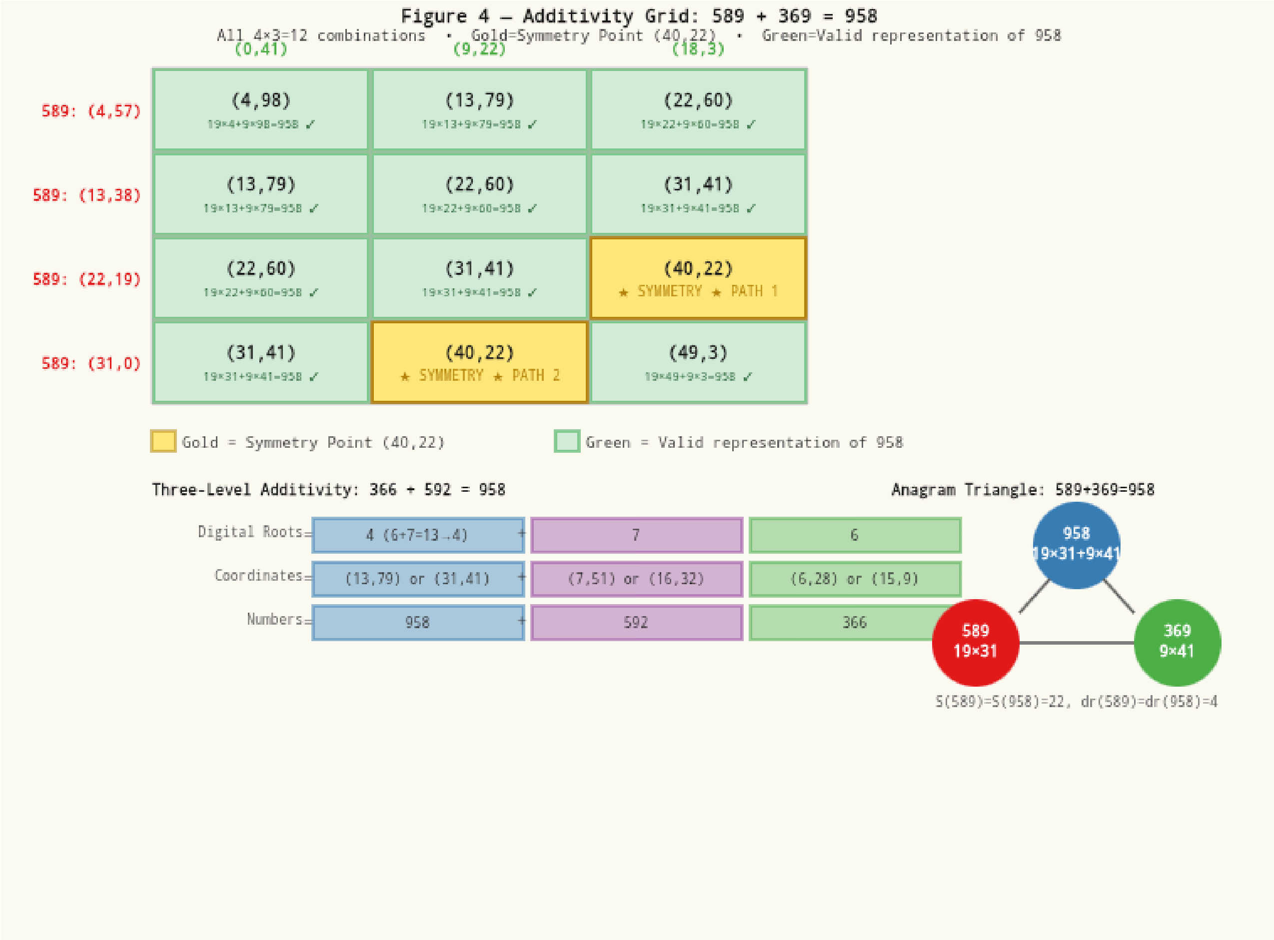


*Figure 4 — Additivity grid: 589+369=958. All 4×3=12 combinations shown. Gold = symmetry point (40,22). Green = valid representation of 958. Three-level additivity and anagram triangle shown below.*

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## §8 Complete Matryoshka Proof (4 Steps + Empirical Verification)

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Every value in the reduction chain of $N$ — intermediate stops and the digital root — appears as an $A$-coordinate in the representations of $N$. The structure scales without loss: however large $N$ may be.

> THEOREM 8.1 — MATRYOSHKA THEORY (COMPLETE PROOF)
>
> Let $N \geq pq$ and let $N \to c_1 \to c_2 \to \ldots \to c_k = \mathrm{dr}(N)$ be the complete reduction chain. Then for every $c_i$ in the chain: $c_i$ is an $A$-coordinate in the representations of $N$, provided $N \geq p \cdot c_i$.

STEP 1 — THE CONGRUENCE INVARIANT

**Claim:** At every step in the chain, $c_i \equiv N \pmod{q}$.

**Base case:** $N$ itself. $N \equiv N \pmod{q}$. ✓

**Inductive step:** Assume $c_i \equiv N \pmod{q}$. Then $c_{i+1} = S(c_i)$, the digit sum of $c_i$. Since $10 \equiv 1 \pmod{9}$, every power $10^j \equiv 1 \pmod{9}$, hence $S(m) \equiv m \pmod{9}$ for every positive integer $m$. Since $q \mid 9$: $S(c_i) \equiv c_i \pmod{q}$. Therefore $c_{i+1} \equiv c_i \equiv N \pmod{q}$. By induction: $c_i \equiv N \pmod{q}$ for all $i$. ✓

STEP 2 — B IS NON-NEGATIVE AND AN INTEGER

Set $A = c_i$, then $B = (N - p \cdot c_i)/q$. From Step 1: $p \cdot c_i \equiv c_i \equiv N \pmod{q}$, so $N - p \cdot c_i \equiv 0 \pmod{q}$. Hence $B$ is an integer. ✓ Non-negativity: we require $N \geq p \cdot c_i$, as stated in the theorem. ✓

STEP 3 — THE REPRESENTATION IS VALID

$p \cdot c_i + q \cdot B = p \cdot c_i + q \cdot (N - p \cdot c_i)/q = p \cdot c_i + N - p \cdot c_i = N$ ✓

STEP 4 — EVERY $C_I$ LIES IN THE A-SEQUENCE

From Step 1: $c_i \equiv N \pmod{q} \equiv A_0 \pmod{q}$. Therefore $c_i = A_0 + k \cdot q$ for some integer $k \geq 0$, and $c_i \leq \lfloor N/p \rfloor$ under the stated condition. Thus $c_i$ falls precisely within the $A$-sequence. ✓ ∎

***Corollary 8.2 — Entropy Observation:*** *dr(N) is the most compressed representation of N — minimal information, but always recoverable as the smallest A-coordinate. The reduction chain maps the complete information path from N to dr(N). No information is lost — it is redistributed across the A-coordinates.*

## Empirical verification

| N | CHAIN | CHAIN VALUES | PRESENT? |
|---|---|---|---|
| 366 | 366→15→6 | 15, 6 | A=15: 15×19+9×9=366 ✓ \| A=6: 6×19+28×9=366 ✓ |

| | | | |
|---|---|---|---|
| 958 | 958→22→4 | 22, 4 | A=22: 22×19+60×9=958 ✓ \| A=4: 4×19+98×9=958 ✓ |
| 9999 | 9999→36→9 | 36, 9 | A=36: 36×19+1035×9=9999 ✓ |
| 19999 | 19999→37→10→1 | 37, 10, 1 | All present ✓ |
| 999999 | 999999→54→9 | 54, 9 | A=54: 54×19+110997×9=999999 ✓ |

### Three-layer case: N = 19999

| LAYER | VALUE | ROLE | REPRESENTATION | VERIFICATION |
|---|---|---|---|---|
| Outer | 19999 | The number | — | — |
| Middle 1 | 37 | First intermediate stop | A=37, B=2144 | 37×19+2144×9=19999 ✓ |
| Middle 2 | 10 | Second intermediate stop | A=10, B=2201 | 10×19+2201×9=19999 ✓ |
| Inner | 1 | Digital root — min. A | A=1, B=2220 | 1×19+2220×9=19999 ✓ |

## §9 Doubling: N = (p + 2q) · A

THEOREM 9.1 — DOUBLING

$B = kA$ if and only if $N = (p + kq)\cdot A$. For $k = 2$: $N = 37A$.

| N | N/37 = A | B = 2A | VERIFICATION |
|---|---|---|---|
| 37 | 1 | 2 | 1×19+2×9=37 ✓ |
| 74 | 2 | 4 | 2×19+4×9=74 ✓ |
| 370 | 10 | 20 | 10×19+20×9=370 ✓ |
| **592** | **16** | **32** | **16×19+32×9=592 ✓** |

| | | | |
|---|---|---|---|
| 740 | 20 | 40 | 20×19+40×9=740 ✓ |

**Observation for (16, 32):** 19×16 − 9×32 = 304 − 288 = 16 = $A$. The difference between the two terms equals the $A$-coefficient. Moreover: $S(288)$ = 2+8+8 = 18 = φ(19).

## §10 Perfect Numbers in the 19-9 System

THEOREM 10.1 — DIGITAL ROOT OF PERFECT NUMBERS

Every even perfect number $P \geq 28$ satisfies $P \equiv 1 \pmod 9$. Consequently $dr_9(P) = 1$ and $A_0 = 1$. Moreover $S(P)$ is a valid $A$-coordinate whenever $P \geq 19 \cdot S(P)$.

PROOF

Powers of 2 modulo 9 have period 6: 2, 4, 8, 7, 5, 1, ... For prime exponents $n \equiv 1$ or $n \equiv 5 \pmod 6$: $P \equiv 1 \pmod 9$, so $dr(P) = 1$ and $A_0 = 1$. □

| N | P | S(P)=A | CHAIN OF A | 19×A | B |
|---|---|---|---|---|---|
| 3 | 28 | 10 | 10→1 ✓ | 190 | 1 (unique: 19+9=28) |
| 5 | 496 | 19 | 19→10→1 ✓ | 361 | 15 |
| 7 | 8128 | 19 | 19→10→1 ✓ | 361 | 863 |
| 13 | 33,550,336 | 28 | 28→10→1 ✓ | 532 | 3,727,756 |
| 17 | 8,589,869,056 | 64 | 64→10→1 ✓ | 1,216 | ~9.5×10⁸ |

### Match observation: S(P) = k + 9

For $p = 7$ and $p = 13$ the digit sums of the two factors satisfy $S(2^{p-1}) = S(2^p-1) = k$, and in both cases $S(P) = k+9$. Both matching primes satisfy $p \equiv 1 \pmod 6$. Whether this congruence is necessary or sufficient for the match is not proved.

| P | $S(2^{N-1})$ | $S(2^N-1)$ | MATCH? | S(P) | Δ |
|---|---|---|---|---|---|
| 5 | 7 | 4 | no | 19 | +8 |
| **7** | **10** | **10** | **YES ✓** | **19** | **k+9=19 ✓** |
| **13** | **19** | **19** | **YES ✓** | **28** | **k+9=28 ✓** |
| 17 | 25 | 13 | no | 64 | +26 |
| 19 | 19 | 28 | no | 55 | +8 |

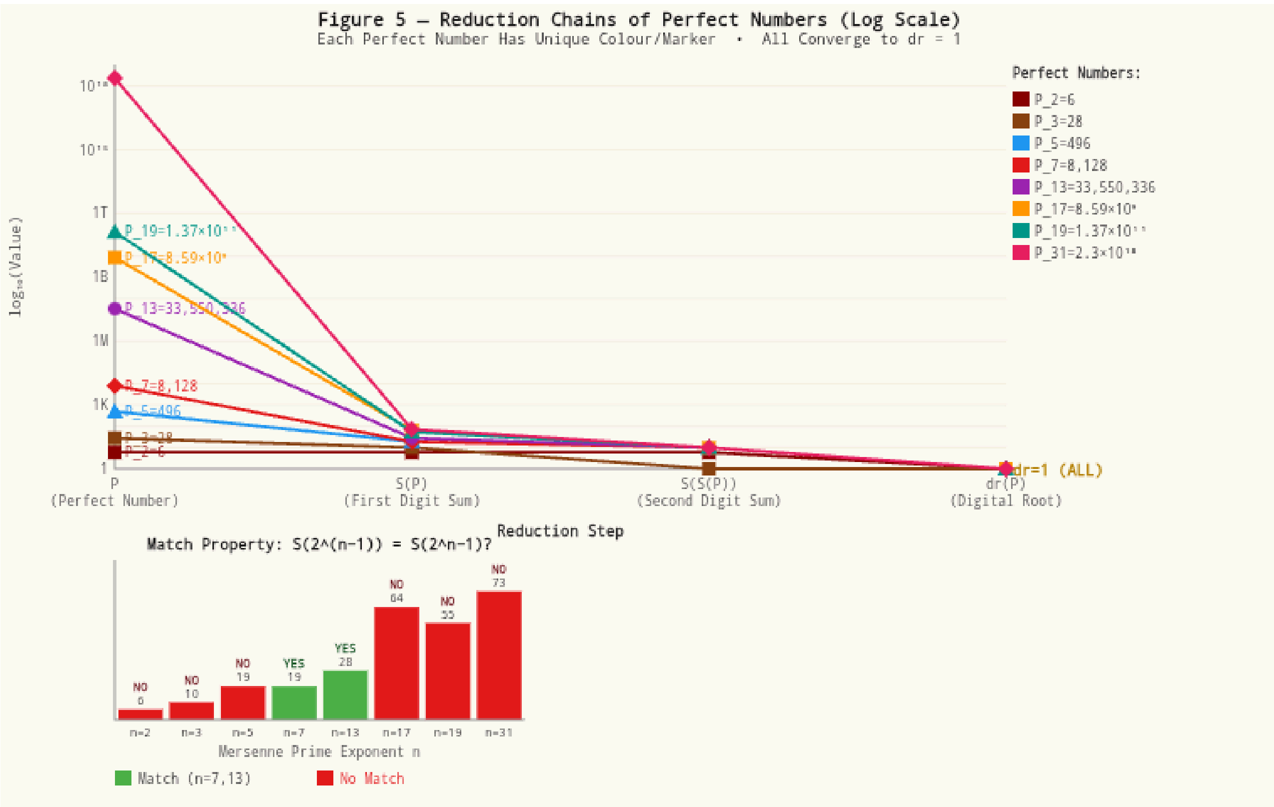


*Figure 5 — Reduction chains of perfect numbers (log scale). Each perfect number has unique colour/marker. All converge to dr=1. Match property S(2^(n−1))=S(2^n−1) shown for each Mersenne prime exponent.*

## §11 Comparative Analysis: Uniqueness of (19, 9)

Systematic scan over all primes $p \leq 200$. Only (19, 9) scores maximally on all five criteria.

| PROPERTY | (19,9) | (37,18) | (199,99) | (331,165) | OTHER P≤200 |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| $\varphi(p)=2q$ | ✓ | ✓ | ✓ | ✓ | ✓ all primes |
| p+q perfect? | ✓ (28) | ✗ (55) | ✗ (298) | ✓ (496) | ✗ |
| Int. stop as A | 12 | 7 | 11 | 5 | 0–11 |
| Max carry-window | 10 | 3 | 10 | 1 | ≤7 |
| A−9=B−19=S(N) | ✓ (12) | ✓ (6) | not tested | ✗ | rarely |
| Closed triangles | 5 | 1 | 0 | 0 | ≤1 |
| $q^2/p$ ratio | 4.26 | 8.76 | 49.25 | 82.25 | variable |

**Key insight:** The combination of all five structural properties is unique to (19, 9). No other pair $(p, q)$ with $p \le 200$ simultaneously satisfies the perfect sum, Euler key, carry-window of length $q$, balance set, and closed triangle conditions.

---

## §12 Classification of Pairs (Classes I–IV)

| CLASS | DEFINITION | EXAMPLES |
|---|---|---|
| **I** | $p \equiv 1 \pmod q$, $\gcd(p,q)=1$ | (10,9), (7,3), (13,4), (19,9) |
| **II** | Class I + $\varphi(p)=2q$ | (7,3), (19,9) |
| **III** | Class II + $p+q$ is an even perfect number | (19,9), (331,165), (5419,2709) |
| **IV** | Class III + closed triangle + full Matryoshka structure | **Only (19, 9)** |

THEOREM 12.1 — FORMULA FOR CLASS-III PAIRS

If $(p, q)$ is a Class-III pair with perfect number $P$: $p = (2P+1)/3$ and $q = (P-1)/3$.

PROOF

From $\varphi(p) = p-1 = 2q$ it follows that $q = (p-1)/2$. Substituting into $p+q = P$ gives $(3p-1)/2 = P$, hence $p = (2P+1)/3$. □

| P | P=(2P+1)/3 | Q=(P-1)/3 | PRIME? | CLASS |
|---|---|---|---|---|
| 6 | 13/3 | 5/3 | — | — |
| **28** | **19** | **9** | ✓ | **IV** |
| 496 | 331 | 165 | ✓ | III |
| 8128 | 5419 | 2709 | ✓ | III |
| 33,550,336 | 22,367,557 | 11,183,778 | ✗ | — |

## §13 Open Problems

V1 — UNIQUENESS OF (19, 9) OPEN

**Statement:** (19, 9) is the only Class-IV pair.

The asymptotic argument shows that $q^2/p$ grows as $P/6$ for Class-III pairs, making the complete carry-window structure structurally impossible for every perfect number greater than 28. The parametric formula $p = (2P+1)/3$ connects the existence of Class-III pairs to Mersenne primes. For every known perfect number beyond $P = 28$, the ratio $q^2/p$ exceeds 8, which eliminates the carry-window property that distinguishes Class-IV.

**Status:** Open — asymptotic support. A complete proof would require either resolving the Mersenne prime conjecture or proving that no carry-window of length $q$ can exist when $q^2/p > 5$.

V2 — FIFTH PERFECT SUM PAIR OPEN

**Statement:** Does there exist an even perfect number $P > 33{,}550{,}336$ for which $p = (2P+1)/3$ is a prime number?

The three known Class-III pairs beyond (19, 9) are (331, 165) with $P = 496$, and (5419, 2709) with $P = 8128$. The next would require $P = 33{,}550{,}336$ (the fifth perfect number), giving $p = 22{,}367{,}557$ — which is not prime. The question is therefore whether any Mersenne prime exponent $n$ beyond those already known produces a prime $p = (2P_n+1)/3$.

**Status:** Open — directly linked to the Mersenne prime problem. No even perfect number beyond 8128 is known to yield a prime $p = (2P+1)/3$.

## §14 Conclusions

We have developed a complete theory for linear Diophantine systems $N = pA + qB$ with $p \equiv 1 \pmod{q}$. The generalised digital root gives the minimal coefficient, the number of representations follows a simple closed formula, and the threshold values lie on arithmetic progressions with common difference $pq$.

The Additivity Theory shows that if $a + b = c$, representations sum simultaneously on three levels: numbers, coordinates, and digital roots. This holds universally — for every triple.

The Matryoshka Theory shows that the complete reduction chain of every $N$ is encoded in its representations. Every intermediate stop and dr($N$) appear as $A$-coordinates. The structure scales without loss.

The second Class-III pair (331, 165) was discovered and analysed. Although it satisfies the perfect sum and the Euler key, it possesses no closed triangle due to the excessively large $q^2/p$ ratio. The asymptotic argument establishes that this ratio grows as $P/6$ for Class-III pairs, making the carry-window property structurally impossible for all perfect numbers greater than 28.

The parametric formula $p = (2P+1)/3$ links the existence of such pairs to Mersenne primes. With this, the mathematical identity of (19, 9) as the unique Class-IV pair is established.

> *"Count, calculate, observe — no interpretations beyond what the data supports."*

APPENDIX A

# Python Code (All 3 Layers)

Consistent, working Python code for the 19-9 system. For large $N$ (roughly $N > 200$) the direct approach is considerably more efficient than the extended Euclidean algorithm.

## Layer 1 — General version (any pair with p ≡ 1 mod q)

```
def a0_general(n, p, q):
    assert p % q == 1
    return n % q

def r_general(n, p, q):
    a0 = n % q
    fm = n // p
    if a0 > fm:
        return 0
    return (fm - a0) // q + 1

def all_representations_general(n, p, q):
    a0 = n % q
    result = []
    a = a0
    while a <= n // p:
        result.append((a, (n - p * a) // q))
        a += q
    return result
```

## Layer 2 — Specific to the 19-9 system (p=19, q=9)

```python
def digital_root(n):
    if n == 0:
        return 0
    return 1 + (n - 1) % 9

def r(n, p=19, q=9):
    a0 = n % q
    fm = n // p
    if a0 > fm:
        return 0
    return (fm - a0) // q + 1

def all_representations(n, p=19, q=9):
    a0 = n % q
    result = []
    a = a0
    while a <= n // p:
        result.append((a, (n - p * a) // q))
        a += q
    return result

def symmetry_point(n, p=19, q=9):
    r_val = digital_root(n)
    for a, b in all_representations(n, p, q):
        if digital_root(a) == r_val and digital_root(b) == r_val:
            return (a, b)
    return None

def g(k, p=19, q=9):
    return p * q * (k - 1) + (p - 1) * (q - 1)

# Verification
assert r(958) == 6
assert r(6666) == 39
assert symmetry_point(958) == (40, 22)
assert symmetry_point(6666) == (24, 690)
assert g(10) == 1683
```

## Layer 3 — Complete Matryoshka calculator

```python
def digit_sum(n):
    return sum(int(c) for c in str(n))

def digital_root(n):
    while n >= 10:
        n = digit_sum(n)
    return n

def reduction_chain(n):
    chain = [n]
    current = n
    seen = {n}
    while current >= 10:
        nxt = digit_sum(current)
        if nxt not in seen:
            chain.append(nxt)
            seen.add(nxt)
        current = nxt
    return chain

def all_representations(n, p=19, q=9):
    if n < 0:
        return []
    a0 = n % q
    result = []
    a = a0
    while a <= n // p:
        b = (n - p * a) // q
        if p * a + q * b == n and b >= 0:
            result.append((a, b))
        a += q
    return result

def link_chain_to_representations(n, p=19, q=9):
    chain = reduction_chain(n)
    reps = all_representations(n, p, q)
    a_values = {a for a, _ in reps}
    coupling = []
    for ci in chain:
        if ci in a_values:
            b = next(b for a, b in reps if a == ci)
            ver = p * ci + q * b
            coupling.append({
                'ci': ci, 'present': True, 'B': b,
                'verification': ver, 'correct': ver == n,
            })
        else:
            coupling.append({
                'ci': ci, 'present': False, 'B': None,
                'verification': None, 'correct': False,
            })
```

```python
    dr_n = digital_root(n)
    symmetry_points = [
        (a, b) for a, b in reps
        if digital_root(a) == dr_n and digital_root(b) == dr_n
    ]
    a0 = n % q
    r_formula = (n // p - a0) // q + 1 if a0 <= n // p else 0
    return {
        'N': n, 'p': p, 'q': q, 'dr_N': dr_n, 'A0': a0,
        'chain': chain, 'chain_str': '→'.join(str(c) for c in chain),
        'R': len(reps), 'R_formula': r_formula,
        'representations': reps, 'coupling': coupling,
        'symmetry_points': symmetry_points,
    }
```

APPENDIX B

# Balance Set A-9 = B-19 = S(N)

All numbers satisfying $A-9 = B-19 = S(N)$. The first ten form an arithmetic progression with common difference $28 = p+q$, starting at $N = 510$. Each step of 28 in $N$ increases both $A$ and $B$ by 1.

| N | A | B | S(N) | A-9 | B-19 | DR(N) |
|---|---|---|---|---|---|---|
| 510 | 15 | 25 | 6 | 6 | 6 | 6 |
| 538 | 16 | 26 | 7 | 7 | 7 | 7 |
| 566 | 17 | 27 | 8 | 8 | 8 | 8 |
| 594 | 18 | 28 | 9 | 9 | 9 | 9 |
| 622 | 19 | 29 | 10 | 10 | 10 | 1 |
| 650 | 20 | 30 | 11 | 11 | 11 | 2 |
| 678 | 21 | 31 | 12 | 12 | 12 | 3 |
| 706 | 22 | 32 | 13 | 13 | 13 | 4 |
| 734 | 23 | 33 | 14 | 14 | 14 | 5 |

| | | | | | | |
|---|---|---|---|---|---|---|
| 762 | 24 | 34 | 15 | 15 | 15 | 6 |
| 790 | 25 | 35 | 16 | 16 | 16 | 7 |
| 818 | 26 | 36 | 17 | 17 | 17 | 8 |
| 846 | 27 | 37 | 18 | 18 | 18 | 9 |
| 874 | 28 | 38 | 19 | 19 | 19 | 1 |
| **958** | **31** | **41** | **22** | **22** | **22** | **4** |
| 986 | 32 | 42 | 23 | 23 | 23 | 5 |

**Structural observation:** The main progression 510, 538, ..., 874 advances in steps of $\Delta = 28 = p+q$. Each step of 28 in $N$ raises both $A$ and $B$ by 1, preserving the balance condition. The gap between $N = 874$ and $N = 958$ marks a structural break.

APPENDIX C

# Complete Family 1675–1683

The nine consecutive integers 1675–1683 form the first contiguous sequence in which each of the nine digital roots 1–9 appears exactly once and every member has reached its full representation count. From $N = 1683$ onward, every integer has at least 10 representations. $N = 1682$ has only 9 representations because DR8 structurally reaches the boundary one step later.

| N | CHAIN | DR(N) | $A_0$ | R(N) | SYMMETRY POINT(S) |
|---|---|---|---|---|---|
| 1675 | 1675→19→10→1 | 1 | 1 | 10 | (28,127) |
| 1676 | 1676→20→2 | 2 | 2 | 10 | (2,182) and (83,11) |
| 1677 | 1677→21→3 | 3 | 3 | 10 | (57,66) |
| 1678 | 1678→22→4 | 4 | 4 | 10 | (31,121) |
| 1679 | 1679→23→5 | 5 | 5 | 10 | (5,176) and (86,5) |
| 1680 | 1680→15→6 | 6 | 6 | 10 | (60,60) ← A=B |

| | | | | | |
|---|---|---|---|---|---|
| 1681 | 1681→16→7 | 7 | 7 | 10 | (34,115) |
| 1682 | 1682→17→8 | 8 | 8 | 9 (!) | (8,170) |
| 1683 | 1683→18→9 | 9 | 0 | 10 | (63,54) |

## All (A,B)-pairs — ★ = Symmetry Point

| N | DR | R | ALL PAIRS |
|---|---|---|---|
| 1675 | 1 | 10 | (1,184) (10,165) (19,146) ★(28,127) (37,108) (46,89) (55,70) (64,51) (73,32) (82,13) |
| 1676 | 2 | 10 | ★(2,182) (11,163) (20,144) (29,125) (38,106) (47,87) (56,68) (65,49) (74,30) ★(83,11) |
| 1677 | 3 | 10 | (3,180) (12,161) (21,142) (30,123) (39,104) (48,85) ★(57,66) (66,47) (75,28) (84,9) |
| 1678 | 4 | 10 | (4,178) (13,159) (22,140) ★(31,121) (40,102) (49,83) (58,64) (67,45) (76,26) (85,7) |
| 1679 | 5 | 10 | ★(5,176) (14,157) (23,138) (32,119) (41,100) (50,81) (59,62) (68,43) (77,24) ★(86,5) |
| 1680 | 6 | 10 | (6,174) (15,155) (24,136) (33,117) (42,98) (51,79) ★(60,60) (69,41) (78,22) (87,3) |
| 1681 | 7 | 10 | (7,172) (16,153) (25,134) ★(34,115) (43,96) (52,77) (61,58) (70,39) (79,20) (88,1) |
| 1682 | 8 | 9 | ★(8,170) (17,151) (26,132) (35,113) (44,94) (53,75) (62,56) (71,37) (80,18) |
| 1683 | 9 | 10 | (0,187) (9,168) (18,149) (27,130) (36,111) (45,92) (54,73) ★(63,54) (72,35) (81,16) |

APPENDIX D

# Unresolved Observations on Perfect Numbers

The following data is presented as observation, not as theorem. The match property $S(2^{p-1}) = S(2^p-1)$ holds for $p = 7$ and $p = 13$, both satisfying $p \equiv 1 \pmod 6$. Whether this congruence is necessary or sufficient has not been proved. The value $\Delta = +8$ recurs at $p = 5$ and $p = 19$; whether this pattern continues is unknown. No closed formula for $S(P)$ in non-match cases has been found.

| P | S($2^{N-1}$) | S($2^N$-1) | MATCH? | S(P) | Δ | NOTE |
|---|---|---|---|---|---|---|
| 2 | 2 | 3 | no | 6 | +1 | P=6, dr=6 (exception) |
| 3 | 4 | 7 | no | 10 | -1 | P=28 |
| 5 | 7 | 4 | no | 19 | +8 | P=496 |
| **7** | **10** | **10** | **YES ✓** | **19** | — | **k+9=10+9=19 ✓** |
| **13** | **19** | **19** | **YES ✓** | **28** | — | **k+9=19+9=28 ✓** |
| 17 | 25 | 13 | no | 64 | +26 | |
| 19 | 19 | 28 | no | 55 | +8 | |
| 31 | 37 | 46 | no | 73 | -10 | |

APPENDIX E

# N = 2520

$$N = 2520 = lcm(1, 2, ..., 10)$$

- dr(2520) = 9, R(2520) = 15
- Two symmetry points: (9, 261) and (90, 90)
- 2520 = 90 × 28 = 90 × (19+9) = 19×90 + 9×90
- Representation (90, 90): A = B — dr(90) = 9 = dr(2520)

Numbers appearing together in a computation do not cause each other. The mathematics here is real and verifiable: 2520 is the smallest positive integer divisible by every integer from 1 to 10, and its position in the 19-9 system follows entirely from its digital root (dr = 9) and its size.

APPENDIX F

# Fibonacci and the Golden Ratio

The golden ratio $\varphi = (1+\sqrt{5})/2 \approx 1.61803398...$ appears in this system via the structural numbers 366, 592 and 958: $366\times\varphi \approx 592.1$ and $592\times\varphi \approx 957.9$. The ratio $592/366 \approx 1.61749$ deviates from $\varphi$ by only $2\times10^{-4}$.

The Pisano period of Fibonacci modulo 19 is 18, and F(18) = 2584 = 19×136 (so B = 0). For every pair in the table, the intermediate stop of each Fibonacci number is a valid A-coordinate in its representations.

| $F_N$ | $F_{N+1}$ | RATIO | DEV. FROM Φ | INT. STOP $F_N$ | INT. STOP $F_{N+1}$ |
|---|---|---|---|---|---|
| **366** | **592** | **1.61748634** | **0.00054765** | **15** | **16** |
| 610 | 987 | 1.61803279 | 0.00000120 | 7 | 24 |
| 987 | 1597 | 1.61803279 | 0.00000046 | 24 | 22 |
| 1597 | 2584 | 1.61803279 | 0.00000018 | 22 | 19 |
| 2584 | 4181 | 1.61803279 | 0.00000007 | 19 | 14 |

**Observation:** For every pair in this table the intermediate stop of each Fibonacci number is a valid A-coordinate in its representations. The ratio approaches φ, but that is a property of the Fibonacci sequence itself — the (19,9)-system makes it visible.

## Convergence towards φ starting from 366 and 592

If one takes 366 and 592 as starting values and repeatedly adds the last two terms — exactly as in the Fibonacci sequence — the ratio of consecutive terms converges towards φ. Starting from 592/366, each step brings the ratio closer.

The second step already produces 958, the third structural number: 366 + 592 = 958. This is not a coincidence: 958 is precisely the number whose digital root, representations, and symmetry point are central throughout this document.

| STEP | N(I-1) | N(I) | RATIO | ΔΦ |
|---|---|---|---|---|
| **1** | **366** | **592** | **1.61748634** | **0.00054765** |

| | | | | |
|---|---|---|---|---|
| **2** | **592** | **958** | **1.61824324** | **0.00020925** |
| 3 | 958 | 1550 | 1.61795407 | 0.00007992 |
| 4 | 1550 | 2508 | 1.61806452 | 0.00003053 |
| 5 | 2508 | 4058 | 1.61802233 | 0.00001166 |
| 6 | 4058 | 6566 | 1.61803844 | 0.00000445 |
| 7 | 6566 | 10624 | 1.61803229 | 0.00000170 |
| 8 | 10624 | 17190 | 1.61803464 | 0.00000065 |
| 9 | 17190 | 27814 | 1.61803374 | 0.00000025 |
| 10 | 27814 | 45004 | 1.61803408 | 0.00000009 |
| 15 | 308462 | 499102 | 1.61803398875 | 0.00000000077 |
| 30 | 420742394 | 680775494 | 1.61803398874989 | 0.000000000000000 |

## When is it a perfect match?

The ratio never reaches φ exactly. $\varphi = (1+\sqrt{5})/2$ is irrational — it cannot be expressed as a ratio of two integers. Every step in the sequence produces a ratio of two whole numbers, and therefore always deviates from φ by some amount, however small.

What the table shows is convergence: each step oscillates above and below φ, with the deviation shrinking rapidly. Within the precision of a computer (15 decimal places), the ratio equals φ from step 30 onward — at that point the numbers involved exceed 400 million. Mathematically, the sequence converges to φ in the limit, but never arrives.

This is precisely the same behaviour as the standard Fibonacci sequence. The structural numbers 366 and 592 are not Fibonacci numbers themselves, but once taken as starting values they generate a sequence with identical convergence behaviour — because the ratio of any two consecutive terms of any such additive sequence converges to φ, regardless of the starting values.

APPENDIX G

# Systematic Scan for p ≤ 200

Selected primes $p \le 200$ with $q = (p-1)/2$. MaxWin = maximum carry-window length.

| P | Q | Q²/P | INT.STOP=A | B=2A COUNT | COUNT A-9=B-19=S(N) | MAXWIN |
|---|---|---|---|---|---|---|
| 3 | 1 | 0.33 | 1 | 1 | 5 | 1 |
| 7 | 3 | 1.29 | 4 | 3 | 0 | 1 |
| 13 | 6 | 2.77 | 3 | 3 | 0 | 1 |
| **19** | **9** | **4.26** | **12** | **15** | **12** | **10** |
| 37 | 18 | 8.76 | 7 | 8 | 6 | 3 |
| 73 | 36 | 17.75 | 11 | 8 | 0 | 6 |
| 109 | 54 | 26.75 | 10 | 12 | 0 | 7 |
| 127 | 63 | 31.25 | 10 | 11 | 0 | 7 |
| 199 | 99 | 49.25 | 11 | 9 | 0 | 10 |

APPENDIX S

# All Theorems and Formulas

## S.1 — Definitions

| LABEL | DEFINITION |
|---|---|
| S.1.1 | $N=pA+qB$, $p\equiv1 \pmod q$, $\gcd(p,q)=1$, $p>q\ge1$, $A,B\ge0$. $R(p,q;N)$ = number of non-negative solutions. |
| S.1.2 | $\mathrm{dr}_q(N) = 1+((N-1) \bmod q)$. For $q=9$: ordinary decimal digital root. |

| | | |
|---|---|---|
| S.1.3 | Reduction chain: $N \to c_1 \to c_2 \to \dots \to c_k = \mathrm{dr}(N)$. $c_{i+1} = S(c_i)$, ends at $c_k \in \{1,\dots,9\}$. | |
| S.1.4 | Symmetry point: $\mathrm{dr}_q(A) = \mathrm{dr}_q(B) = \mathrm{dr}_q(N)$. | |
| S.1.5 | $\mathrm{G}(p,q;k) = pq(k-1) + (p-1)(q-1)$. For (19,9): G(1)=144, G(10)=1683. | |
| S.1.6 | $\mathrm{T}(t,r) = pr + pq(t-1)$. Rows $+pq = 171$; columns $+p = 19$. | |

## S.2 — Main Theorems

| LABEL | FORMULA / STATEMENT | REF. |
|---|---|---|
| S.2.1 | $A_0 = N \bmod q = \mathrm{dr}_q(N)$ for $N \geq pq$ | §2 |
| S.2.2 | $\mathrm{R}(p,q;N) = \lfloor (\lfloor N/p \rfloor - \mathrm{dr}_q(N))/q \rfloor + 1$ · Asymp.: $\mathrm{R}(N) \sim N/(pq)$ | §3 |
| S.2.3 | $\mathrm{G}(k) = 171k - 27$ [for (19,9)]. Every $N \geq \mathrm{G}(k)$ has $\geq k$ representations. | §4 |

## S.3 — Structural Theorems

| LABEL | STATEMENT | REF. |
|---|---|---|
| S.3.1 | Additivity: if $a+b=c$ then $(A_1+A_2, B_1+B_2)$ represents $c$. Levels: numbers, coordinates, digital roots all add simultaneously. | §7 |
| S.3.2 | Matryoshka: for $N \geq pq$ with chain $N \to c_1 \to \dots \to c_k = \mathrm{dr}(N)$: every $c_i$ is an A-coordinate of $N$, provided $N \geq p \cdot c_i$. | §8 |
| S.3.3 | Entropy corollary: $\mathrm{dr}(N)$ is the innermost layer. The reduction chain is the information path from $N$ to $\mathrm{dr}(N)$. No information is lost. | §8 |

## S.4 — Additional Theorems

| LABEL | STATEMENT | REF. |
|---|---|---|
| S.4.1 | $B = kA \iff N = (p+kq) \cdot A$. For $k=2$: $N = 37A$. | §9 |
| S.4.2 | Every even perfect $P \geq 28$: $P \equiv 1 \pmod 9$, $\mathrm{dr}(P) = 1$, $A_0 = 1$. $S(P)$ is always a valid A-coordinate. | §10 |

| | | |
|---|---|---|
| S.4.3 | If $(p,q)$ is Class-III with perfect $P$: $p=(2P+1)/3$, $q=(P-1)/3$. | §12 |
| S.4.4 | Match property: if $S(2^{p-1})=S(2^p-1)=k$ then $S(P)=k+9$. Verified for $p=7$ and $p=13$. | §10 |
| S.4.5 | Anchor family: $N$=1675–1683 each different dr (1–9), R($N$)=10, except $N$=1682 (R=9). From $N$=1683: every $N$ has ≥10 representations. | App. C |

## S.5 — Formulas on one page

| FORMULA | DESCRIPTION | REF. |
|---|---|---|
| $A_0 = N \bmod q = \mathrm{dr}_q(N)$ | Smallest A — the main theorem | S.2.1 |
| $B_0 = (N-p\cdot A_0)/q$ | Corresponding B at $A_0$ | S.2.1 |
| $A \to A+q,\ B \to B-p$ | All representations via one step | §1 |
| $\mathrm{R}(N)=\lfloor(\lfloor N/p\rfloor-A_0)/q\rfloor+1$ | Number of representations | S.2.2 |
| $\mathrm{R}(N)\sim N/(pq)$ | Asymptotic growth | S.2.2 |
| $\mathrm{T}(t,r)=pr+pq(t-1)$ | Threshold values per digital root | S.1.6 |
| $\mathrm{G}(k)=pq(k-1)+(p-1)(q-1)$ | Family boundary = min. N with R≥k | S.2.3 |
| $\mathrm{G}(k)=171k-27$ [(19,9)] | Family boundary in the 19-9 system | S.2.3 |
| $N=(p+kq)\cdot A \Leftrightarrow B=kA$ | Doubling condition | S.4.1 |
| $\mathrm{dr}(P)=1$ for all even $P\geq 28$ | Perfect numbers | S.4.2 |
| $p=(2P+1)/3,\ q=(P-1)/3$ | Class-III pair at perfect P | S.4.3 |
| $A-9=B-19=S(N)$ | Balance condition | App. B |

## S.6 — Open Problems

| NR. | CONJECTURE | STATUS |
|---|---|---|
| V1 | (19,9) is the only Class-IV pair | OPEN asymptotic support |
| V2 | Even perfect P > 33,550,336 with (2P+1)/3 prime? | OPEN Mersenne question |